Customer Churn Prediction and Business Insights Report

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Project: End-to-End Customer Churn Prediction Pipeline

Tools Used: Python, Pandas, Scikit-learn, Power BI, VS Code, Jupyter Notebook

Model Used: Random Forest Classifier

Model Accuracy: 78.32%

Project Objective:

The objective of this project is to develop an end-to-end machine learning pipeline to predict customer churn using the Telco Customer Churn dataset. The project includes data extraction, transformation, and loading (ETL), exploratory data analysis (EDA), feature engineering, machine learning model development, model evaluation, and the creation of an interactive Power BI dashboard. The primary goal is to identify customers who are likely to leave the company and provide actionable business insights that can help improve customer retention and support data-driven decision-making.

Dataset Overview:

The project uses the **Telco Customer Churn** dataset, which contains customer demographic information, account details, subscribed services, billing information, and churn status. After the ETL process and data cleaning, the dataset contained **7,032 customer records** and **21**

features. The target variable is **Churn**, which indicates whether a customer has left the company's services (Yes) or continues as an active customer (No). This dataset is widely used for customer retention analysis and churn prediction problems.

ETL (Extract, Transform, Load) Process:

An ETL pipeline was developed in Python to automate the data preparation process before model training and dashboard creation. The pipeline follows three major stages:

Extract: The raw customer churn dataset was loaded from a CSV file using the Pandas library.

Transform: The dataset was cleaned by handling missing values in the **TotalCharges** column, converting data types, removing incomplete records, and preparing the data for further analysis.

Load: The cleaned dataset was saved as a new CSV file (cleaned_customer_churn.csv), which was later used for machine learning model development and Power BI dashboard creation.

ETL Workflow:

1. Loaded the raw Telco Customer Churn dataset.
2. Checked data types and missing values.
3. Converted TotalCharges to numeric format.
4. Removed rows containing missing values.
5. Verified the cleaned dataset.
6. Saved the cleaned dataset for further analysis.

Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) was performed to understand customer behavior, identify important patterns, and discover factors influencing customer churn. Various visualizations were created using Python libraries such as Matplotlib and Seaborn to analyze relationships between customer attributes and churn status. The analysis helped identify the most influential features before training the machine learning model.

Key Findings:

- Customers with **month-to-month contracts** showed the highest churn rate.
- Customers using **fiber optic internet service** experienced higher churn compared to DSL users.
- **Electronic check** was associated with a relatively higher churn rate than other payment methods.
- Customers with **higher monthly charges** were more likely to churn.
- Customers with **longer tenure** were generally less likely to leave the company.
- Gender had **minimal impact** on customer churn.

Visualizations Used:

Visualization	Purpose
Churn Distribution	Understand churn proportion
Contract vs Churn	Analyze churn by contract type
Internet Service vs Churn	Compare churn across internet services
Payment Method vs Churn	Study payment-related churn
Correlation Heatmap	Identify feature relationships
Feature Importance	Understand influential variables

Machine Learning Model:

A Random Forest Classifier was selected to build the customer churn prediction model due to its strong performance on classification tasks and ability to handle both numerical and categorical features. Before training, categorical variables were encoded using Label Encoding, and the dataset was divided into training and testing sets using an 80:20 ratio. The model was trained on the processed data and evaluated using standard classification metrics.

Model Summary:

Parameter	Value
Algorithm	Random Forest Classifier
Train-Test Split	80% - 20%
Target Variable	Churn
Accuracy	78.32%
Problem Type	Binary Classification

Model Evaluation:

The trained model achieved an overall accuracy of **78.32%**, demonstrating good predictive performance for identifying customers who are likely to churn. A confusion matrix and classification report were generated to evaluate the model's performance in terms of correctly classified customers, precision, recall, and F1-score. The results indicate that the model can effectively support customer retention strategies by identifying high-risk customers.

Power BI Dashboard Summary:

An interactive Power BI dashboard was developed to visualize customer churn trends and support business decision-making. The dashboard provides an overview of customer retention metrics, churn distribution, and the relationship between churn and various customer attributes. Interactive slicers allow users to filter the dashboard dynamically, making it easier to analyze customer behavior across different segments.

Dashboard Components:

Component	**Description**
KPI Cards	Total Customers, Churned Customers, Retained Customers, Churn Rate, Average Monthly Charges, Average Tenure
Donut Chart	Customer Churn Distribution
Column Chart	Contract Type vs Customer Churn
Bar Chart	Internet Service vs Customer Churn
Column Chart	Payment Method vs Customer Churn
Column Chart	Gender vs Customer Churn
Column Chart	Average Monthly Charges by Churn
Interactive Filters	Contract, Internet Service, Gender, Payment Method

Dashboard Benefits:

- Provides a quick overview of customer churn.
- Enables interactive filtering for deeper analysis.
- Highlights key business factors influencing churn.
- Supports data-driven business decisions.
- Improves understanding of customer retention trends.

Key Business Insights:

1. **Customers with month-to-month contracts showed the highest churn rate**, indicating that customers without long-term commitments are more likely to leave.
2. **Fiber optic internet users experienced higher churn** compared to DSL users, suggesting that service quality, pricing, or customer expectations may require further investigation.
3. **Customers paying through electronic checks exhibited relatively higher churn**, indicating that payment methods may influence customer retention.
4. **Customers with higher monthly charges were more likely to churn**, highlighting pricing as an important factor affecting customer decisions.
5. **Customers with longer tenure were significantly less likely to churn**, demonstrating that long-term customer relationships improve retention.
6. **The overall churn rate was approximately 26.6%**, meaning that nearly one out of every four customers discontinued the service.

Business Recommendations:

1. Introduce attractive offers and loyalty programs for customers with month-to-month contracts to encourage long-term subscriptions.

2. Review the pricing and service quality of fiber optic internet plans to improve customer satisfaction.
3. Provide personalized retention campaigns for customers identified as high churn risks by the machine learning model.
4. Offer discounts or bundled packages to customers with higher monthly charges to reduce the likelihood of churn.
5. Encourage customers to switch to annual or two-year contracts through promotional pricing and additional benefits.
6. Continuously monitor churn trends using the Power BI dashboard to support timely business decisions.

Conclusion:

This project successfully demonstrated the complete development of an end-to-end customer churn prediction pipeline, beginning with data extraction and preprocessing through ETL, followed by exploratory data analysis, machine learning model development, and interactive dashboard creation. The Random Forest Classifier achieved an accuracy of **78.32%**, providing reliable predictions for customer churn. The Power BI dashboard and business insights generated during this project can help organizations identify customers at risk of leaving and support data-driven strategies to improve customer retention. Overall, this project highlights the value of combining data engineering, machine learning, and business intelligence to solve real-world business problems.